

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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Abstract

Using high-frequency administrative data from ADP, we document six facts characterizing labor market shifts following the widespread adoption of generative AI. Early-career workers (ages 22-25) in AI-exposed occupations experienced 16% relative employment declines, controlling for firm-level shocks, while employment for experienced workers remained stable. Adjustments occur primarily via employment rather than compensation, with employment changes concentrated in occupations where AI automates rather than augments labor. Results are robust to excluding technology firms and occupations that are remotable. These six facts provide early large-scale evidence consistent with generative AI disproportionately impacting entry-level workers in the American labor market.

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¶Latest version: <https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/>

1 Introduction

The proliferation of generative artificial intelligence (AI) has sparked a global debate about its potential impact on the labor market. This discourse spans utopian predictions of enhanced productivity, dystopian fears of widespread job displacement, and skeptical views that AI will have minimal effects on employment or productivity. Historically, technologies have affected different tasks, occupations, and industries in different ways, replacing work in some, augmenting others, and transforming still others. These heterogeneous effects suggest that there may be “canaries in the coal mine” which are harbingers of more widespread effects of AI.

Recent discussion coincides with rapid improvements in AI capabilities and adoption. From 2023 to 2024, AI systems improved from solving 4.4% of coding problems to 71.7% on SWE-Bench, a widely used benchmark for software engineering (Maslej et al., 2025). AI has also improved in areas like language understanding, subject knowledge, and reasoning. A recent paper found that current systems could match or outperform up to 47 percent of industry professionals on a pre-defined benchmark of economically valuable tasks (Patwardhan et al., 2025). At the same time, AI systems are increasingly widely adopted. Hartley et al. (2025) find that LLM adoption at work among U.S. survey respondents above age 18 reached 46% by June/July 2025.¹

Given the better capabilities and widespread adoption, a central concern, amplified in recent headlines, is whether AI is beginning to supplant human labor, particularly for younger, entry-level workers in highly exposed professions like software engineering and customer service.² For instance, in May of 2025, Dario Amodei, co-founder and CEO of Anthropic predicted that AI could wipe out roughly 50 percent of all entry-level white-collar jobs within five years (Morris, 2025).

Despite the intensity of this debate, empirical evidence has struggled to keep pace with technological advancement, leaving many fundamental questions unanswered. This paper confronts this empirical gap by leveraging a large-scale, high-frequency administrative dataset from ADP, the largest payroll software provider in the United States. Our sample consists of monthly, individual-level payroll records through September 2025, encompassing millions of workers across tens of thousands of firms. This rich panel structure allows us to track employment dynamics with a high

¹Similarly, Bick et al. (2024) found that in late 2024, nearly 40% of the U.S. population age 18-64 reported using generative AI, with 23% using it for work weekly and 9% daily.

²Improved productivity of workers in an occupation could lead to either reduced or increased employment, depending on, among other things, how elastic demand is for the output of those workers.

degree of granularity, providing a near real-time view of labor market adjustments. By linking this data to established measures of occupational AI exposure and other variables, we can quantify the realized employment changes since the widespread adoption of generative AI.

This paper systematically presents six key facts that emerge from the data, offering an assessment of how the AI revolution is reshaping the American workforce.

Our first key finding is that we uncover substantial declines in employment for early-career workers (ages 22-25) in occupations most exposed to AI, such as software developers and customer service representatives. In contrast, employment trends for more experienced workers in the same occupations, and workers of all ages in less-exposed occupations such as nursing aides, have remained stable or continued to grow.

Our second key fact is that overall employment continues to grow robustly, but employment growth for young workers has been stagnant since late 2022. In jobs less exposed to AI, young workers have experienced comparable employment growth to older workers. In contrast, workers aged 22 to 25 have experienced a 6% decline in employment from late 2022 to September 2025 in the most AI-exposed occupations, compared to a 6-9% increase for older workers. These results suggest that declining employment in AI-exposed jobs drives stagnant overall employment growth for 22- to 25-year-olds.

Our third key fact is that not all uses of AI are associated with declines in employment. In particular, entry-level employment has declined in applications of AI that *automate* work, but not those that most *augment* it. We distinguish between automation and augmentation empirically using estimates of the extent to which observed queries to Claude, the LLM, substitute or complement the tasks in that occupation. While we find employment declines for young workers in occupations where AI primarily automates work, we find employment *growth* in occupations in which AI use is most augmentative. These findings are consistent with automative uses of AI substituting for labor while augmentative uses do not.

Fourth, we find that employment declines for young, AI-exposed workers remain after conditioning on firm-time effects. One class of explanations for our patterns is that they may be driven by industry- or firm-level shocks such as interest rate changes that correlate with sorting patterns by age and measured AI exposure. We test for a class of such confounders by controlling for firm-time effects in an event study regression, absorbing aggregate firm shocks that impact all workers at a

firm regardless of AI exposure. For workers aged 22-25, we find a 15 log-point decline in relative employment for the most AI-exposed quintiles compared to the least exposed quintile, a large and statistically significant effect. Estimates for other age groups are much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

Fifth, the labor market adjustments are visible in employment more than in compensation. In contrast to our findings for employment, we find little difference in annual salary trends by age or exposure quintile, suggesting possible wage stickiness. If so, AI may have larger effects on employment than on wages, at least initially, or even that AI may boost wages for as many workers as it hurts.

Sixth, the above facts are largely consistent across sample constructions designed to address various alternative explanations for the core findings. Our results are not driven solely by computer occupations or by occupations susceptible to remote work and outsourcing. We also find that the AI exposure taxonomy did not meaningfully predict employment outcomes for young workers further back in time, before the widespread use of LLMs, including during the unemployment spike driven by the COVID-19 pandemic. The patterns we observe in the data appear most acutely starting in late 2022 and early 2023, around the time of rapid proliferation of generative AI tools.³ They hold for both occupations with a high share of college graduates and ones with a low college share, suggesting deteriorating education outcomes during COVID-19 do not drive our results. For non-college workers, we find evidence that experience may serve as less of a buffer to labor market disruption, as low college share occupations exhibit divergent employment outcomes by AI exposure up to age 40.

While we explore a variety of alternative explanations, we caution that the facts we document may in part be influenced by factors other than generative AI. Taken as a whole, our results are consistent with the hypothesis that generative AI has begun to affect entry-level employment. We intend to continue to track the data on an ongoing basis to assess whether these trends change in the future.

³In particular, OpenAI introduced ChatGPT in November 2022. It was reported to have over 100 million active users by January 2023 and 1.7 billion website visitors in October 2023 (DeVon, 2023).